

0 2

A student investigated the effect of three types of disinfectant on the growth of *Lactobacillus* bacteria.

During the investigation, the student:

- boiled the agar before pouring the agar plates
- transferred 0.5 cm^3 of a diluted liquid culture of *Lactobacillus* onto each agar plate
- left some agar plates as controls
- added to other agar plates different concentrations of the disinfectants as shown in **Table 1** on page 5.

After 2 days, she counted the number of colonies of bacteria on each agar plate.

0 2 . 1

Explain the purpose of:

[2 marks]

boiling the agar _____

transferring the same volume of liquid culture onto each agar plate.



The three disinfectants used by the student were Lysol, propan-2-ol and ammonia.

Table 1 shows the student's results.

Table 1

Concentration of disinfectant / arbitrary units	Number of colonies of bacteria		
	Lysol	Propan-2-ol	Ammonia
0	300	300	300
5	0	290	300
10	0	195	295
15	0	0	275
20	0	0	240

0 2 . 2

The liquid culture the student transferred was diluted by 1 in 10 000 (10^{-4}).

Use information in this question to calculate how many bacteria were present in 1 cm^3 of undiluted liquid culture.

[2 marks]

Answer = _____

Question 2 continues on the next page

Turn over ►



0	2	.	3
---	---	---	---

The student concluded that the minimum concentration of propan-2-ol needed to stop the growth of *Lactobacillus* was 15 units. This conclusion is incorrect.

Describe how you could obtain a more accurate estimate of the minimum concentration of propan-2-ol needed to stop the growth of this species of bacterium.

[2 marks]

6

